Operations for Autonomous Spacecraft

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***Abstract*—Onboard autonomy technologies such as planning and scheduling, identification of scientific targets, and content-based data summarization, will lead to exciting new space science missions. However, the challenge of *operating* missions with such onboard autonomous capabilities has not been studied to a level of detail sufficient for consideration in mission concepts. These autonomy capabilities will require changes to current operations processes, practices, and tools. We have developed a case study to assess the changes needed to enable operators and scientists to operate an autonomous spacecraft by adopting a common model between the ground personnel and the onboard algorithms. We assess the new operations tools and workflows necessary to enable operators and scientists to convey their desired intent to the spacecraft, and to be able to reconstruct and explain the decisions made onboard and the state of the spacecraft. Mock-ups of these tools were used in a user study to understand the effectiveness of the processes and tools in enabling a shared framework of understanding, and the ability of the operators and scientists to effectively achieve mission science objectives.**

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1. INTRODUCTION

Advanced onboard autonomy capabilities including autonomous fault management (e.g., [1], [2]), planning, scheduling (e.g., [3]), and execution (e.g., [4]), selection of scientific targets (e.g., [5]), and onboard data summarization and compression (e.g., [6]) are being developed for future space missions. These autonomy technologies hold promise to enable missions that cannot be achieved with traditional ground-in-the-loop operations cycles due to communication constraints, such as high latency and limited bandwidth, combined with dynamic environmental conditions or limited mission lifetime. Classes of missions enabled by autonomy include in situ and subsurface exploration of icy giant moons, coordinated deep space fleet missions, and fast flybys in which changing features or a lack of a priori knowledge of

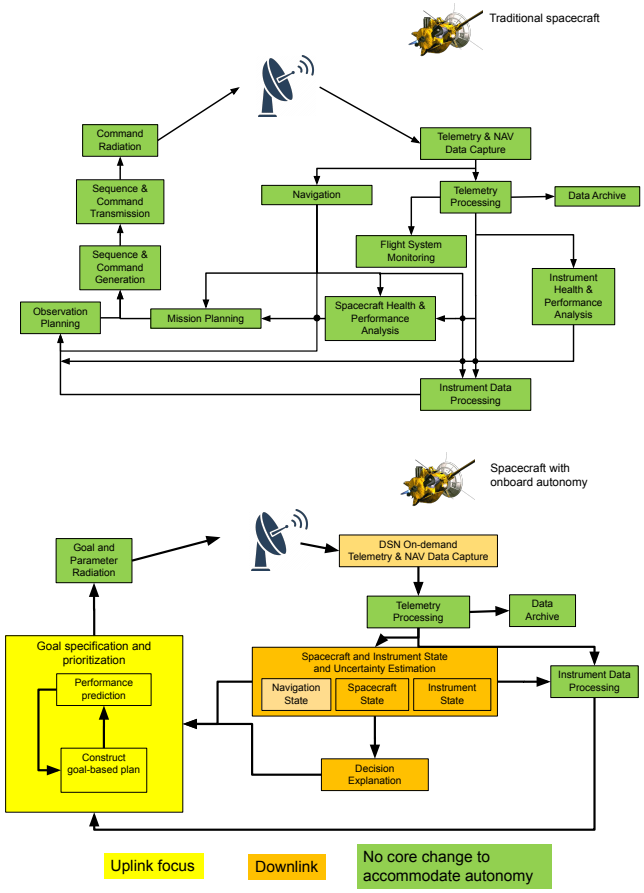

Figure 1. Traditional operations workflow (top) and operations workflow for future autonomous spacecraft (bottom). Future spaceflight missions are likely to employ advanced onboard autonomy capabilities; in order to successfully operate such missions, new workflows and software tools will be needed to provide *intent* to the onboard autonomy, determine *why* autonomy made its decisions, and assess the spacecraft *state*.

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position requires fail-operational capability and autonomous detection and pointing. Onboard autonomy capabilities can also increase science return, improve spacecraft reliability, and have the potential to reduce operation costs. As a compelling example, autonomy has already significantly increased the capabilities of Mars rover missions, enabling them to perform autonomous long-distance navigation and autonomous data collection on new science targets [5], [7].

While there has been a focus on the development of onboard autonomy capabilities, the challenges of *operating* a deep space spacecraft with these autonomous capabilities and the impact on ground operations has never been studied to a level of detail sufficient for consideration in mission concepts. To enable scientists and engineers to operate autonomous spacecraft, new operations tools and workflows must be developed (Figure 1). In this paper, we study the problem of *operations of autonomous spacecraft* and, specifically, we identify workflows and software tools that are well-suited for this problem. We then apply these workflows and tools to a realistic mission concept representative of a future "ice giant" outer planet (e.g., Neptune) tour mission.

At a conceptual level, uplink teams must communicate their science and engineering *intent* to onboard planning, execution and autonomous science software, and assess the likely impact of such intent on the spacecraft state. Downlink teams, in turn, must reconstruct and explain what decisions were made by autonomy, assess the spacecraft's state (which is strongly influenced by onboard autonomy), and identify anomalies that may otherwise be hidden by autonomous decisions.

The uplink workflow we propose leverages modeling technology that facilitates an iterative design process of operational intent, including capturing intent and constructing plans with that intent. We focus on workflows for outcome/execution prediction, explanation, as well as advisory techniques (e.g., "to fix undesirable behavior, add/change this constraint"), to facilitate the operators' learning process, while helping reassure them that the spacecraft will fulfill the desired intents and complete the plan successfully.

The proposed downlink workflow focuses on two thrusts. The first thrust is spacecraft state estimation and propagation, a challenging problem in presence of onboard autonomy, which may alter the spacecraft state in response to information that is not immediately available on the ground. The second thrust is explanation of the decisions taken by onboard autonomy. User interfaces must indicate what decisions were made by the autonomy algorithms and relate why the decisions were made to the intent specified by ground operators, the spacecraft state (including possible anomalies), and the perceived state of the environment.

We evaluate the proposed tools through a user study set in a realistic mission scenario inspired by a notional ice giant tour mission (science rich, but power and telecommunication limited). The tools, workflows, and lessons learned will directly inform future science and exploration missions across a variety of mission classes, including surface missions (e.g., Europa and other Icy World Lander, Mars Sample Return, and Venus Lander), small body exploration (e.g., fast flybys, Centaur rendezvous), and other ambitious mission types.

Contribution

The contribution of this paper is fourfold.

First, to motivate the investigation into operations of autonomous spacecraft, we identify three classes of onboard science autonomy applications, and a corresponding set of eight science scenarios in the context of a notional mission to the Neptune-Triton system (including both nominal and off-nominal situations), that are enabled by onboard autonomy capabilities and that are likely to challenge current operations paradigms.

Second, we assess current operations workflows, and identify changes, including new roles and new tools, that are likely to be required as more autonomy is introduced in deep space missions.

Third, we propose a set of user interface designs and software tools designed to address the challenges of operations for autonomy, providing operators with the ability to express their intent to onboard planners and to assess the spacecraft's state and the autonomy's decisions.

Finally, we assess the performance of the tools in a user study with JPL operators, showing that the proposed tools and workflows are able to support high-level workflows representative of operations of autonomous spacecraft, and identifying directions for future research.

Organization

The rest of this paper is organized as follows. In Section 2, we describe the mission and operations concept under consideration in this work, and identify a set of autonomy-enabled mission scenarios that challenge current operations paradigms. In Section 3, we describe current operations uplink and downlink workflows, and investigate how the workflows and roles will have to be adapted to accommodate onboard autonomy. Section 4 presents a suite of user interfaces designed to support uplink and downlink operators, and Section 5 presents software tools developed to provide data to the user interface (UX) tools. Section 6 reports the preliminary results of a user study that assessed the suitability and performance of the proposed user interfaces with JPL spacecraft operators and scientists. Finally, in Section 7, we draw our conclusions and lay out directions for future research.

2. OPS CONCEPT AND MISSION CASE STUDY

We focus on a concept of operations for a notional spacecraft exploring the Neptune-Triton system. The selection of a specific mission concept provides a concrete setting in which to develop and exercise tools and workflows; the Neptune-Triton system is an especially interesting setting because the significant light-speed latency, low available bandwidth, short duration of flybys, and dynamic scientific phenomena make autonomy highly attractive to fulfill primary mission objectives, but also make operations of such a mission very challenging.

The notional mission concept was informed by several prior mission concept studies including the Neptune Odyssey mission concept [8], Trident Mission concept [9] and Ice Giants Study [10]. We selected a subset of tour orbits, including several close flybys of Triton, and a subset of representative instruments, specifically, a wide-angle camera, narrow-angle camera, sub-millimeter spectrometer, and plasma and particles instrument. Figure 2 shows the set of orbits considered in the study.

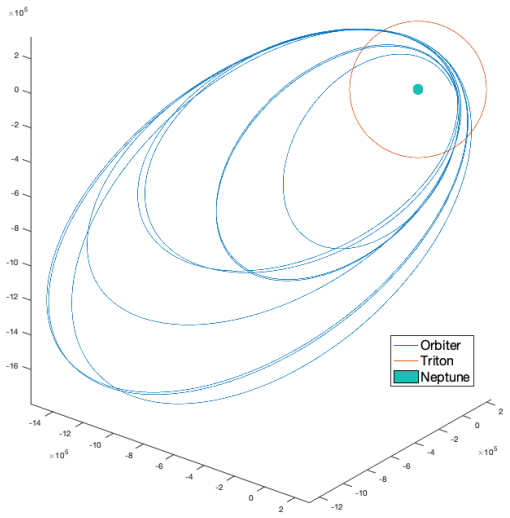

Figure 2. The set of orbits considered in the concept of operations.

Within this notional mission, we identified three classes of science observations and, within these, eight scenarios that exercise a variety of autonomy capabilities, including autonomous science event detection, planning, scheduling and execution, and failure detection, identification, and recovery.

Science campaign classes and scenarios

The scenarios considered in this work can be separated into three broad classes of scientific observations that benefit from onboard autonomy, namely, autonomous monitoring, event-driven opportunistic observations while mapping, and event-driven opportunistic observations during targeted observations.

Monitoring—In monitoring campaigns, an instrument or suite of instruments collects an extended observational data set with the goal of characterizing the behavior of the observed system or natural phenomenon. With onboard autonomy, the data collection campaign is adapted based on observation data. Within the monitoring class, we considered two scenarios, namely:

- *Magnetospheric Variability Detection.* The magnetospheric variability detection scenario exercises *onboard adaptive data compression* in which magnetospheric data readings from the plasma and particles instrument are used by the onboard autonomy to decide whether to store high-frequency, losslessly-compressed readings or low-frequency, binned data (thus leaving more room for other data products) based on the level of magnetospheric activity.
- *Magnetospheric Reconnection Event Detection*. Occasionally, magnetospheric field lines reconnect with each other. As it is difficult to predict these high-science-value events, in this scenario, the onboard autonomy monitors high-frequency data from the plasma and particles instrument, looking for magnetospheric reconnection events. If such an event is detected, the corresponding data is saved in lossless format for downlink; if no such event is detected, the data is discarded.

Event-driven opportunistic observations while mapping—Mapping of a body's surface is typically a pre-planned activity; autonomy can enhance mapping by (i) changing observation parameters on the fly (e.g., camera parameters), (ii) adjusting the schedule in response to unexpected events (e.g., a camera reset), and, crucially, (iii) allowing mapping to be executed in parallel with other *opportunistic* activities, scheduling opportunistic observations for high-value but fleeting events. Within this class of science campaigns, we considered three scenarios, namely:

- *Mapping Triton and Plume Detection*. While executing a pre-planned Triton mapping activity, a plume is detected by the onboard autonomy software. The autonomy algorithms then modify the onboard plan to collect observations of the plume with both cameras as well as the spectrometer, and re-plan the mapping task to minimize the loss of coverage while prioritizing plume observations.
- *Fault Detection, Isolation, and Recovery (FDIR) during mapping*. In this scenario, the camera resets in the midst of executing a pre-planned Triton mapping task. The onboard autonomy recognizes the interruption, ascertains functionality, and re-plans the remaining observations so as to maximize the amount of the target area covered despite the shorter time available for mapping.
- *Mapping Neptune and Storm Detection*. An atmospheric storm is detected while the spacecraft is performing a pre-planned mapping of Neptune. The autonomy reacts by revising the spacecraft plan to collect targeted observations of the storm (whose location may be time-varying and imperfectly known) with its cameras and spectrometer, and re-planning the mapping task to minimize the loss of coverage.

Event-driven opportunistic observations during targeted observations—Similar to mapping, targeted observations are typically a pre-planned activity; autonomy can provide increased science returns by adapting observation parameters, revising observation opportunities if more or less time than expected is available for observations, and, critically, interspersing opportunistic observations with pre-planned observations. We considered three scenarios within this class:

- *Target selection*. The spacecraft is given a ranked list of targets to observe on the surface of Triton. A camera reset causes an observation to take longer than expected. Onboard autonomy then re-plans subsequent observations based on the priorities pre-specified by ground operators, ensuring that once-in-a-mission observations are acquired, possibly at the expense of lower-priority ones.
- *FDIR affects science plan during critical engineering event*. A non-critical camera observation is planned during an engine burn. During the burn, FDIR intervenes to counter an unexpected increase in power usage by interrupting the observation. The interrupted science activity is re-planned for a later time.
- *On-the-fly instrument parameter adjustment*. While imaging a target, autonomy adjusts the narrow-angle camera's observation parameters (specifically, the exposure time and number of exposures to stack) based on the level of noise observed in the images, while ensuring that the resulting observations will fit the downlink budget.

Table 1 reports in detail the uplink and downlink operations capabilities exercised by each scenario. Collectively, the scenarios exercise a number of key capabilities including planning and scheduling, event detection, and FDIR.

In the remainder of this paper, we discuss the tools and workflows that will be used to effectively operate spacecraft in these scenarios.

3. OPERATIONS WORKFLOW

In this section, we describe current JPL operations workflow for missions in a class comparable to the concept of operations under consideration, and highlight areas and roles that will change with the advent of more autonomous missions. Figure 3 provides a high level representation of the new workflow, highlighting key roles and interactions between teams in both uplink and downlink operations.

Uplink Operations

Uplink teams must communicate science and engineering intent to onboard autonomy software and assess the expected impact of such intent on the spacecraft state. The proposed uplink tools leverage previous JPL research on modeling plans to facilitate an iterative design process of science intent, including capturing intent and constructing plans that will execute the intent. We focus on a workflow that includes intent capture/modeling, outcome/execution prediction, explanation of elements in the predicted outcomes (e.g., undesirable performance), as well as advisory techniques (e.g., "to fix undesirable behavior, add/change this constraint"). The proposed workflow aims to facilitate the operators' learning process, while helping reassure them that the spacecraft will achieve the target intents and complete the plan successfully. In what follows, we elaborate on the uplink workflow and a set of supporting tools for intent capture and outcome prediction; tools for explanation and advising will be the subject of future work.

We used the Europa Clipper operations concept as a starting point and identified departures required by increased onboard autonomy. The revised workflow and the supporting framework includes a set of key design principles and progression from uplink to downlink.

In most missions (including, e.g., Europa Clipper and Perseverance Rover), planning occurs on several different time horizons, where the cycle time itself is dependent on the mission. Although the terms for these planning cycles are not consistent across missions, we will refer to a longer cycle as "strategic" and to a shorter cycle as "tactical". The process of capturing intent starts before launch and continues iteratively at the strategic cadence by defining campaigns and initial goals for the different flybys and science opportunities. Intent is updated as necessary at the tactical level based on information arriving from downlink (e.g. spacecraft telemetry, science data). While subsequent planning depends on the development of these initial conditions from downlink, we have not yet addressed this part of the process in detail. Here we will focus on the workflow for tactical planning, where intent is revisited and goals might be updated and/or added for the next flyby (or set of flybys).

At the beginning of the uplink tactical planning cycle for autonomous spacecraft, *scientists*, *engineers* and *operators* go through the process of *intent capture*, starting by revisiting the goals for the next flyby(s) based on the downlink data and analysis. Scientists and engineers then have the opportunity to make changes to the goals/plan while getting instant feedback on the viability of their changes and on their impact on overall mission progress and performance. The viability and impact analysis here is based on an initial, evaluation of a low fidelity assessment of possible onboard autonomy output, e.g., based on the most likely, nominal science and engineering scenario.

Scientists can overload the plan with new observations/goals at this stage, and instead of culling the plan to stay within resource constraints, science negotiations rank science goals by priority, with low priority goals less likely to be executed onboard. The *autonomy engineer* oversees the merging of changes into the main plan. The team then collectively reviews these preliminary "low fidelity" outputs, implements iterations as needed, and approves advancing to the "high fidelity" evaluation of possible outcomes and autonomy output. A high fidelity evaluation is the conducted in the *outcome prediction* phase.

During the outcome prediction phase, simulations produce a more realistic and comprehensive view of the new plan's impact to the projected mission progress and performance. Iterations continue, as the team decides to make minor tweaks or drop problematic goals entirely. The team can inspect individual cases or "clusters" of related cases to understand outcomes that might happen onboard and investigate problematic plans that approach undesired limits. *Explanation* of these problematic cases (either by manually inspecting logs, state history, timelines and traces or by using an automated supporting system) plays an important role by i) providing an understanding of the causes and circumstances in which they occur and ii) updating and adjusting goals to avoid or minimize undesirable outcomes and maximize favorable ones (possible adjustments can be derived by inspection or by also having an automated *advising* system to recommend modifications). When the team converges in this iterative model-predict-adjust process, the new (or updated) set of goals is sent to the spacecraft for onboard planning and execution. Such a model-predict-adjust process is key to help the team build trust in the onboard autonomy since it allows a shared understanding of the autonomous behavior.

Downlink Operations

Downlink teams operating future autonomous spacecraft will be tasked with explaining what decisions the onboard autonomy algorithm made, reconstructing what happened onboard, and identifying anomalies that may otherwise may have gone undetected by the onboard autonomy capabilities. New downlink workflows and tools have been designed that focus on two thrusts. The first thrust is *spacecraft state estimation* and propagation of the spacecraft state (including available energy, temperatures, health of spacecraft subsystems, and consumption of onboard resources). Enabling ground personnel to gain a reliable understanding of an autonomous spacecraft's state (e.g., health, resources, etc.) is a challenging problem, as the onboard autonomy may alter the spacecraft state in response to information that is not immediately available on the ground; in addition, propagating the spacecraft state is critical to providing initial conditions to uplink teams. The second thrust is *explanation* of the decisions taken by onboard autonomy. This is achieved through user interfaces that capture what decisions were made by autonomy, and relate the decisions to (i) the intent provided by ground operators, (ii) the spacecraft state (including possible anomalies), and (iii) the perceived state of the external environment (e.g., events of interest detected by the onboard autonomy).

Current downlink operations concepts—Space flight projects employ teams of downlink engineers to monitor and review spacecraft system status and report the spacecraft's status to uplink planners, as part of a continuous cycle of operations over the entire course of a mission. The specific downlink operations concept of a given mission is strongly influenced not only by the mission type, but also by characteristics such as the available communications bandwidth for spacecraft engineering health and status data (which may be a small per-

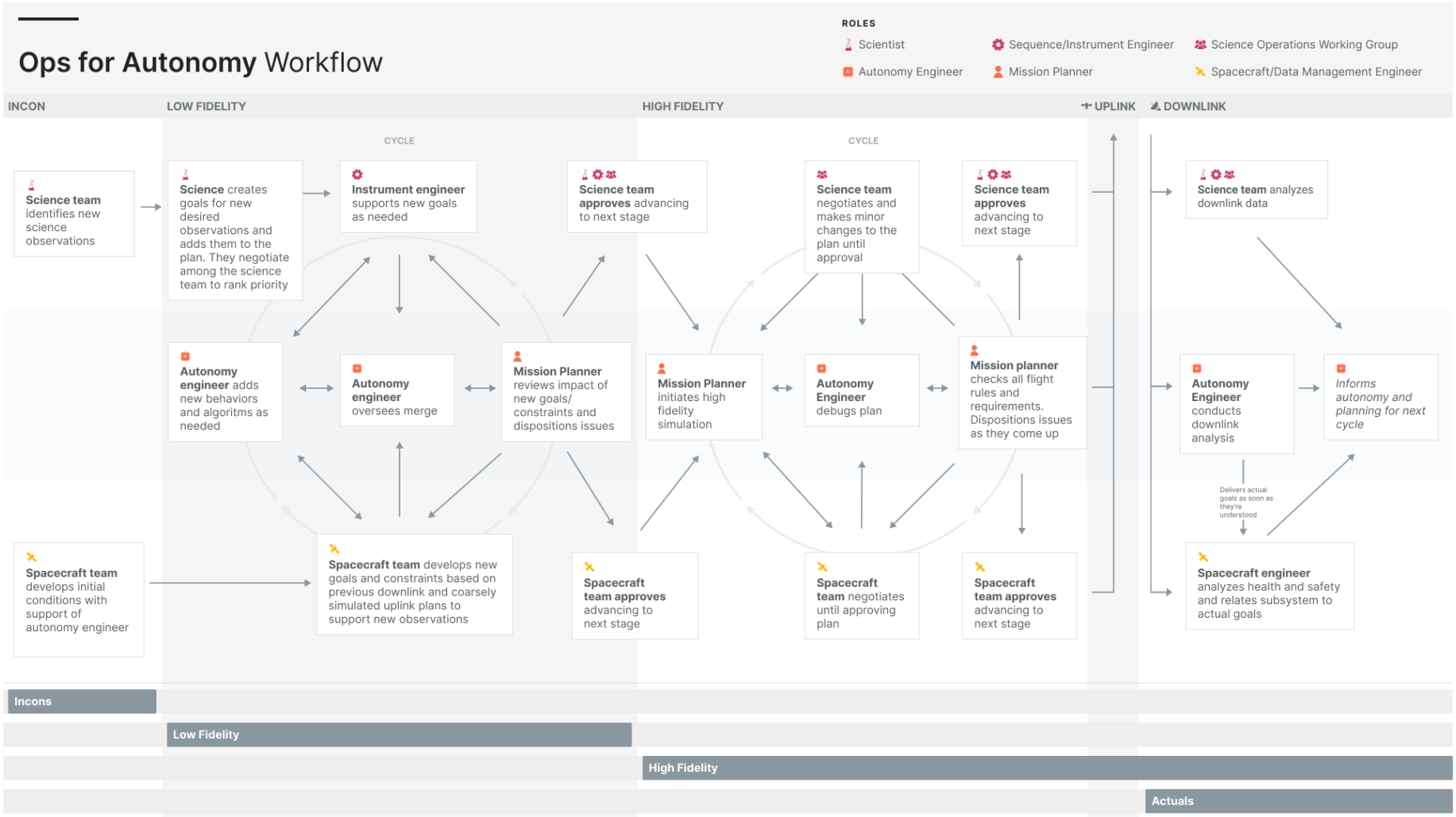

Figure 3. Uplink and Downlink Workflow: Describes the high level operations process. It starts with the development of initial conditions from the previous downlink and identifying new science observations. Then, the team uses low fidelity models to understand the impact and viability of the new goals. When the team accepts the low fidelity predictions, they move to high fidelity modeling, where they evaluate the impact on a larger scale and with more realism. They uplink the output. When the downlinked telemetry ("actuals") arrives, operators compare the actuals against predicts to understand the onboard behavior and new science. Instrument and science teams develop and prioritize science goals to achieve mission science requirements. They maintain models of the environment and instrument behavior in support of ensuring mission success and instrument health and safety. Cross-cutting roles include 1) the mission planner who oversees the integrated plan and dispositions both issues to fix and opportunities to improve it to the appropriate team, and 2) the autonomy engineer who owns the system level autonomous behavior. They support the development of new autonomous behaviors and ensure the performance of the autonomy. The spacecraft team members own the performance of their respective subsystems. They maintain behavior models in support of ensuring spacecraft health and safety.

centage of the total bandwidth available) and the frequency of downlink opportunities. Generally, the biggest driver to the downlink operations process is the timing of receipt of downlink spacecraft engineering telemetry, relative to the need to feed-forward that information into planning operations. Another important driver is the light-speed latency between the spacecraft and the Earth, where long latencies (and the resulting long operational turnaround times) are themselves drivers for increasing use of onboard autonomy.

Certain missions, particularly Earth orbiters, are capable of downlinking with very little latency and also typically are not as restricted in terms of engineering downlink bandwidth. These missions may have styles of downlink operations ranging from hands-on "continuous operations" to "lights out operations" where little operational monitoring is done by people, outside of anomaly response situations.

Deep space (e.g., Mars) landed missions typically have a longer downlink latency, often accompanied by a short turnaround time required to carry forward downlink analysis results and initial conditions into planning. This is especially true of rover missions such as Curiosity and Perseverance, where a large amount of complex data (including both scientific observations and engineering data) must be reviewed in a very short time span, in order for uplink teams to have time to perform all planning functions ahead of the final available uplink of the day. Further, lander missions often depend on orbital relays for delivery of data to Earth, given the higher bandwidth capabilities of orbiter relays compared to direct-to-Earth communications from the landers; this further constrains the downlink operational process timing and cadence to be dependent on the available orbiter relay communications windows. Critically, in anomaly situations, a lander typically just stops whatever it is doing, allowing time for ground operations to address the issue.

Deep-space flyby and multi-flyby missions, such as Cassini, JUNO and Europa Clipper, present all the challenges of deep space surface operations, and in particular the need to examine complex science and engineering data to assess spacecraft health and inform future planning. Deep-space flyby operations introduce an additional key constraint: if an anomaly occurs, there is no option to just "stop flying", and a failure to respond in a very short time span may cause loss of mission.

Impact of autonomy on downlink operations concepts— In general, downlink operations of highly autonomous spacecraft includes *all* of the tasks and challenges associated with non- or low-autonomy missions, but brings with it the additional challenge of understanding whether or not actual spacecraft actions and behaviors are appropriate or anomalous, given a large possible set of valid outcomes. In non- or low-autonomy missions, the *nominal* behavior of the spacecraft is typically unimodal and therefore more straightforward to characterize, given the command sequence provided by uplink; in contrast, in highly autonomous missions, a large number of possible outcomes may all be compatible with the uplink's intent, depending on the spacecraft's state and on its perceived environment.

The impact of increased autonomy on downlink for these different types of missions includes:

- Downlink operations will typically require a high knowledge of uplink intent, both to interpret onboard decisions and to assess the spacecraft's state (specifically, to assess whether the spacecraft's behavior is consistent with the provided intent).
- The presence of autonomy results in a large number of possible outcomes onboard, which cannot be fully determined a priori (indeed, the reason for onboard autonomy is to act on information that is not available on the ground); while prediction tools used by uplink operators can help assess the a priori likelihood and impact of each of these outcomes, operators will have to interpret telemetry from the spacecraft (and compare it to predicted outcomes) in order to fully understand onboard behavior. New tools are likely to be required to compare predictions (which are likely to be highly multi-modal) to actual telemetry, so as to assess the autonomy's behavior and the spacecraft's state.
- Operations are likely to require new types of information related to autonomous behaviors to be downlinked and evaluated by operations (e.g., the inputs and execution traces of autonomy modules). This is a particular area of challenge, as long latency and low bandwidth, which are key drivers to adoption of autonomy, may preclude the ability to downlink large amounts of information related to the behaviors (e.g., images analyzed by an event detector, or the full spacecraft state considered by an onboard planner in each re-planning cycle). This leaves ground operators to decipher the behavior of onboard autonomy with incomplete information; software tools are likely to be required to "fill in the gaps" in the data based on *models* of the spacecraft, its environment, and the onboard autonomy.

A downlink workflow for autonomy— Downlink spacecraft engineers use a variety of software tools to perform their analysis functions for present-day missions. Tools include analysis scripts, reporting systems, and a range of graphical visualizations, including both 2D and 3D views as needed for specific analysis tasks. Of course the primary data being analyzed and visualized is the data downlinked from the spacecraft. Depending on the role of each subsystem engineer, this may include both spacecraft engineering health and status, as well as science data used in engineering analysis (e.g., rover camera images).

There are three general types of spacecraft data used in analysis of JPL missions:

- Time series data representing onboard measurements of spacecraft state over time. JPL missions generally refer to this type of data as "channelized telemetry" or "channels", with each channel representing a time series of measurements from spacecraft hardware sensors, as well as data reported by software components (e.g., onboard memory states). Channelized telemetry is the most widely used type of data, going back to the earliest flight missions, and still plays an important role for current and future missions in the reporting of spacecraft state. However, over time spacecraft engineers found the limitations of pure time series data to be a hindrance to operations, and new data types have emerged over the years.
- Event Records (EVRs) representing single events that occurred onboard the spacecraft. Rather than the single data value of a channel record, each ground EVR record contains a message string, which contains further spacecraft state information embedded in that message.
- Engineering Data Products, each containing a range of types of information, depending on the need. There are a wide variety of data products used by projects, including snapshots of state such as memory and data management states.

All of these data types are packetized for downlink delivery, and typically, during a downlink communications session or soon after, data packets are processed by ground telemetry processing tools. In some cases downlink operators monitor data via "real time" displays which are updated as data is processed by the ground system; in other cases, operators wait until data processing is complete on the ground, and review the overall spacecraft states via reports and graphical dashboards. Downlink constraint checks (often called "alarms") are performed on data both in real time and in post-processing, checking for violations such as data values going over limits (e.g., power levels are too high), invalid behaviors or combinations of behaviors (e.g., activity A cannot ever overlap activity B), and for other types of undesired situations. Further, ground systems often include automated notifications to alert operators of an issue, given there may be a very short turnaround time for mission engineers to respond to the issue and possibly prevent the loss of science data or even the spacecraft itself.

Once data is processed on the ground and in the hands of operators, a variety of tools and techniques are employed to analyze spacecraft health and performance. System status dashboards are used by projects to organize and summarize data by subsystem. Automated and user-triggered reports help to answer specific types of questions and pass forward system status to leads and other teams. Interactive tools are used for exploratory analysis, particularly in response to anomalous or otherwise unexpected behaviors.

Many missions incorporate prediction of onboard state and behavior as part of uplink planning, and operators often compare those predicts against actual telemetry during tactical and strategic analysis of the mission. In most cases, there is only one predict generated for any given system state, making it fairly straightforward to compare an actual against a predict. However with an autonomous system, there might be a large number of possible outcomes, which can greatly complicate the ability of an operator to identify if onboard behaviors are within allowable ranges and expectations or not.

Furthermore, the combination of autonomous behaviors and limited downlink bandwidth create a challenge for operators to understand "why" the autonomous system made certain decisions. Even if all available engineering data could be downlinked, along with science results, to put together the whole story as to what happened onboard, it may still be a challenge to make sense of that data, especially if such understanding needs to happen in a short time span in order to uplink or modify new goals for the spacecraft.

Our work focused on these challenges, and identified two tools to improve downlink operations for autonomous missions: (1) an analysis tool capable of visualizing actual telemetry results against a range of possible predicts in the context of onboard events, and (2) a tool capable of providing the information needed to assess "why" the autonomous component made a given series of decisions.

4. USER INTERFACE TOOLS

We now describe a set of User Interface (UI) designs (implemented as mock-ups) to support the aforementioned iterative operations both for uplink and for downlink operations. We organize them in three groups: Intent Capture (uplink), Outcomes Prediction (uplink), and Downlink Analysis.

Intent Capture

The core of this technology is an intent-oriented hierarchical plan specification, from strategic to tactical, which generally aligns with current mission practices of large ground planning teams at NASA JPL. Building on previous work at JPL [11] [12] [13] [14] we designed a set of UI tools to progressively capture and specify intent as science campaigns. In this work, campaigns are composed of: a constrained set of goals (a desired state value or a high level activity, e.g., survey the magnetosphere, or monitor for plume activity), metrics to evaluate progress toward the goals (i.e., key performance indicators, or KPIs) and their validity range for assessment of execution (e.g., resource usage, frequency of a command cycling due to delays), execution variability to capture uncertainty related to the environment and the spacecraft actuation (e.g., exogenous events, activities run long/short), and relationships between goals (e.g., priorities). Relationships between goals are a critical element to codify, and are not usually explicitly captured on missions, but are rather brought to light through the process of team discussions. Even the simple choice between continuing a mapping task or interrupting it to point an instrument at an unforeseen and valuable opportunistic target requires the spacecraft to have a complex and fully-determined decision logic, and requires operators to consider the possible consequences for the science mission, the spacecraft, and onboard resource budgets.

We specifically designed four UIs to allow scientists, engineers, autonomy engineers and operators to specify campaigns and their respective set of goals: 1) Science Planning tool, 2) Metric Definition tool, 3) Variability Definition tool, and 4) Task Networks Modeling tool.

Science Planning tool—This tool provides a familiar way for scientists and operators to a) search for observation opportunities and b) create and update observation goals and organize them within the context of a set of campaigns for the target mission. The UI design in Figure 4 illustrates the design that allows users to search for science observation opportunities, for example across different upcoming flybys, by specifying observation requirements and constraints. For example, scientists can search for observation opportunities that meet geometrical and imaging parameters, and preview potential conflicts. They can click on targets of interest on a 3D model of the planetary body and inspect the resulting footprint. Once an opportunity is identified as satisfying, users can add the observation as a goal in its corresponding campaign.

The tool also allows the direct specification of goals in the form a set of desired activities (e.g., observation, detection) without necessarily using the search mechanism. Figure 5 shows an example of the creation of two goals (one conditional on execution of the other) to monitor a particular location on the surface of Triton, and perform a follow-on observation if a plume is detected. The figure also shows the geometric element on the right hand side, which is an important assistive visual element for creating goals for observations. During tactical planning, scientists and operators can rapidly make changes to the plan, and get instant feedback on the viability of their changes and their impact on overall mission progress and performance. For viability, the tool runs a surrogate of the onboard planning capability on the ground (in this project we use the MEXEC planning and execution system [4]) in nominal (or most likely) scenarios to check for constraint violations. With respect to impact, the small progress bars to the right of the campaign title shows the impact of that goal in the overall mission compared to the

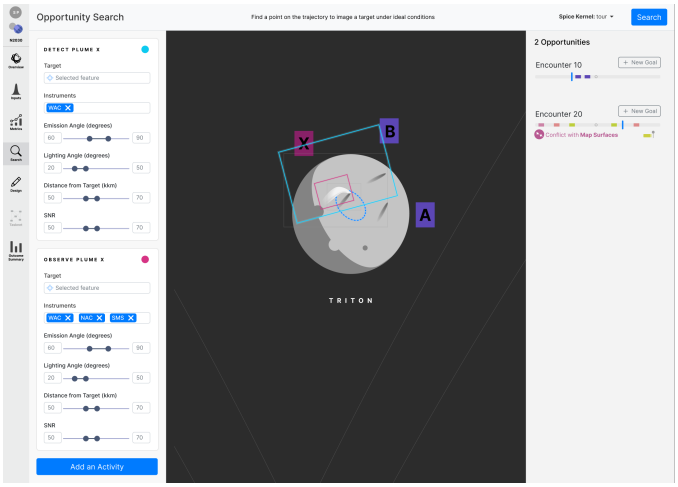

Figure 4. Science Planning: Allows scientists to search for observation opportunities in upcoming flybys

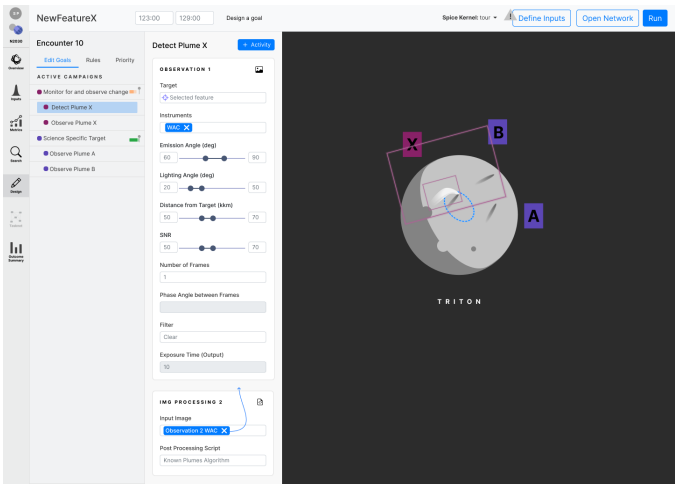

Figure 5. Science Planning: Allows scientists to input campaigns and their respective goals for the mission.

original set of goals.

It is important to note that these tools are domain dependent, meaning that they are designed to support science goal specification for a multi-flyby mission for a single spacecraft. This front end, in particular, is not meant to be used in other domains, e.g., ground vehicles. We intentionally designed it to be a familiar visual presentation for scientists for that particular mission. This helps the user express their intent more naturally, as opposed to requiring them to translate their intents to unfamiliar and unnatural representation languages. In this case, the tools provide appropriate views and interactions depending on the user role (e.g., scientist, operator, etc). All the campaigns and goals are represented and stored in the background in a common language across the different views and tools. This common ground representation is called a Task Network in this work. We will later cover a more domain independent modeling tool for capturing campaigns and goals. That is, operators and scientists could also directly work on the Task Network modeling tool if desired.

Metric Definition—This tool allows users to specify a set of metrics for each campaign, to evaluate the spacecraft's progress and performance toward goals achievements and campaign completion. The tool design provides templates for encoding the metric, e.g., mapping state variables to thresholds or bounds (e.g., the minimal number of plumes that must be detected and observed is 10). This requires users to specify metrics formally and encode the procedure in which progress and completion are evaluated (such as a linear function where one plume observed means progress of 10%, and ten plumes observed achieves 100% completion of that campaign goal). Figure 6 illustrates the UI design for metric specification, showing the "number of plumes" metric as an example. Progress and impact is also shown for each campaign, based on the inputs given in the Mission Planning tool. Metrics are key inputs to evaluate both current and predicted spacecraft performance. Note that the overwhelming majority of metrics are usually captured and specified early in the mission, or at the strategic planning level.

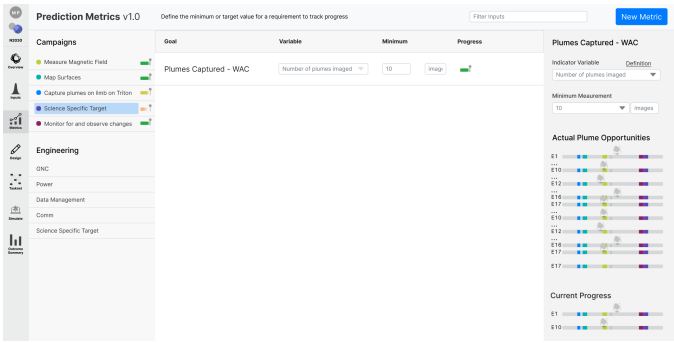

Figure 6. Metric Definition: allows scientists, engineers, autonomy experts and operators to input the key performance indicators that are used to evaluate the expected and actual performance of the spacecraft throughout each flyby and also throughout the mission.

Variability Definition tool—This tool (shown in Figure 7) allows scientists, engineers, autonomy engineer and operators to represent and model uncertainty with respect to a multitude of aspects that might impact the performance of the onboard autonomy while executing its mission. These aspects include *environmental uncertainty and variability*, such as the number of plumes or features on the surface of Triton that might be detected, or *engineering uncertainty and variability* such as the probability that a camera will go into a fault state when in operations, or probability distributions over the duration and power consumption of activities performed onboard. Herein, users represent variability by specifying uncertainty in the form of probability distribution (e.g., Gaussian or Uniform distribution) or discrete percentage. Figure 6 illustrates the variability definition tool. The variability specification is fundamental to study the space of circumstances that the spacecraft might face, and the basis for the prediction phase proposed in this work. It is from these uncertainty models that we sample execution scenarios and evaluate the distribution of possible outcomes.

New additions or updates to variability specification can occur in any stage of the mission. Updates, for example, can come from downlink data, both at the tactical level (e.g. a magnetosphere model might need adjustments), or at the strategic level with trend and history analysis (e.g., the duration of plumes at a certain location can be better estimated given historical data from previous flybys).

Task Network tool—This tool leverages previous JPL work on modeling goals [14] using a formalism called Task Networks (*tasknets* for short). The tool combines the task network

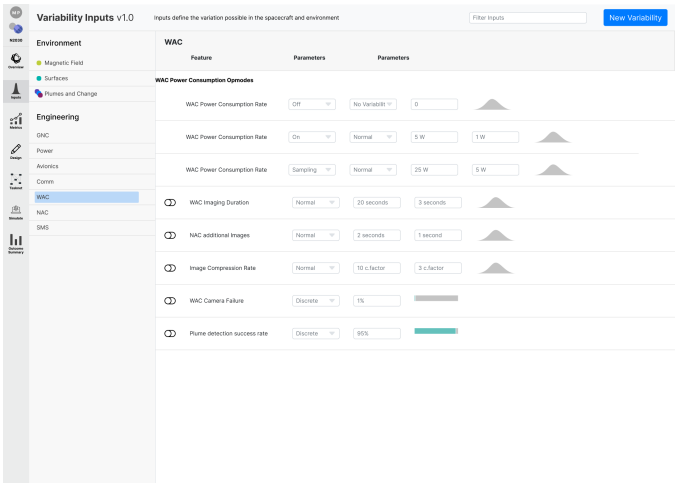

Figure 7. Variability Definition: allows scientists, engineers, autonomy engineers and operators to input uncertainty with respect to a multitude of aspects that might impact the performance of the onboard autonomy spacecraft.

editing and visualization, an output of the predicted task execution on a timeline (for a nominal, or most likely, scenario), and a log of messages including predicted EVRs, onboard planner decisions, and projected constraint violations. Figure 8 shows the UI design for the Task Network tool.

On the tasknet editing and visualization front (graph view on Figure 8), operators can visualize state conditions, state impacts (effects), resource constraints, priority constraints, and ordering constraints across tasks in the task network graph view in order to identify missing and conflicting resources and dependencies, or other issues preventing a task from being scheduled as expected. They can also navigate task hierarchy by zooming out to review the goals included in the plan, or zooming in to a specific goal to focus on lower level tasks. Operators can inspect task details including the commands, priority, and command parameters that belong to a task, along with task authorship history. The tool additionally enables merging tasknets together to combine goals, with capabilities akin to Git version control.

The predicted data shown in the timeline comes from a surrogate of the onboard planner (MEXEC in this work), in order to check if a nominal plan is scheduled as expected. Operators can inspect a moment in time across the different types of data with a brushing capability in order to correlate events, the active task, and resource timelines. Warnings and tooltips draw operators' attention to goals that failed to schedule (or to be achieved), and then it offers the most likely explanations generated by inference (see Section 5 for more details on the inference mechanism) for factors that led to failures as well as nominal outcomes. Operators can also choose to view the final output timeline, or animate individual planning cycles in order to understand how the schedule evolved.

We expect that operators will most frequently generate tasknets using templates and parameters from other supporting tools (e.g., the Science Planning tool), but in other situations, they can manually construct them by pulling in tasks from a template library. In this work, goals from the Science Planning tools are translated to task network elements and added to a master task network, where all the goals are represented, elaborated/expanded and merged. The resulting master task network is the main data product that is uplinked to the spacecraft and handed to the onboard planner. The Task Network tool is central to the intent capture process and will be used both at the strategic and tactical planning levels.

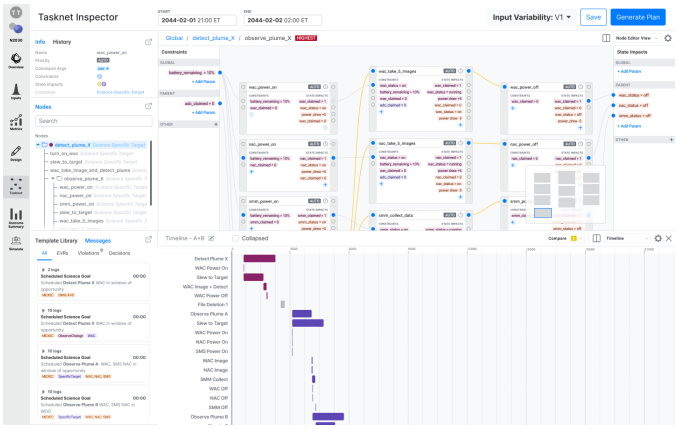

Figure 8. Task Network: allows engineers, autonomy experts, and operators to author and check task networks.

Outcomes Prediction

In order to offer the uplink team a more complete overview of the potential behaviors of the onboard autonomy, we predict the various outcomes that may result from a given task network and given uncertainty models by running an array of high-fidelity simulations. The collected predicted outcomes can then be used by the uplink team to not only observe the expected execution, but also attach confidence values to the various goals and activities within the generated plans. As such, repeated simulation runs and collection of the outcomes fit within the proposed iterative workflow of uplink operations, which ultimately serves the goal of increasing the confidence of the uplink team in the expected behavior and performance of the onboard autonomy with the provided goals.

The software design of the simulation and prediction tool is described in Section 5; in this section, we focus on UI tools that allow users to explore the outcome of the simulations.

Simulation Mission Planning Prediction Review tool—This tool shows the aggregated summary of all the simulation runs [15] for a given task network, as well as the metrics and variability specifications. Figure 9 shows the predicted outcomes (for the target tasknet) on the left-hand side, ordered from most likely to least likely, aiding the operator in more easily deciphering the expected behavior of the constructed plans. The green and red arrows inform the impact of an added goal (in this case, observing Plume X) on the outcome distribution. For example, the percentage of cases in which Observation A and B will be both performed decreased (red arrow pointing down).

The left-hand side panel also presents the option for filtering the output for specific outcomes. Additionally, the main section contains a timeline view showing the scheduled activity times of the science goals, the modes of all instruments, as well as important spacecraft states including as storage usage and battery status. Since this view displays an aggregate of all outcomes, the charts on the timeline view showcase the overlaid results in a gradient-like pattern indicating all the

possible values for the various outcomes. All in all, this view is crucial for the operator in examining all of the possible outcomes after running through the prediction engine.

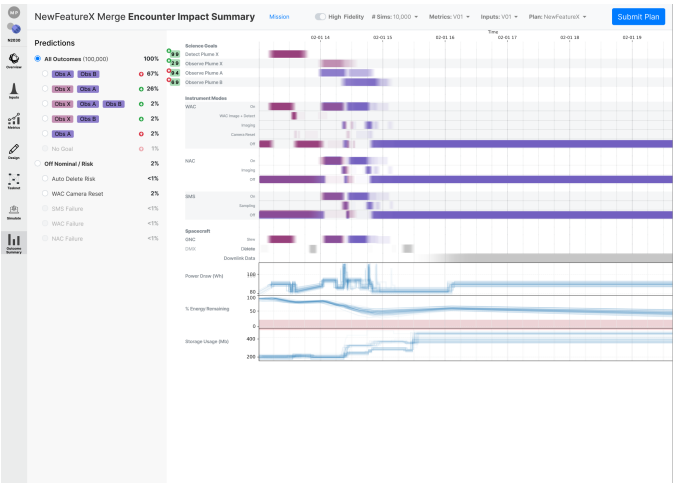

Figure 9. Mission Planning Prediction Results tool: shows the aggregated summary of all simulation runs for a given task network

Mission impact tool—The mission impact UI view (shown in Figure 10) provides an overview of the simulations spanning the whole mission (that is, looking into all flybys), highlighting the impact of newly-added goals to the progress and success of the campaigns and to performance trends. This view also shows how the plans perform with respect to key performance indicators, and the uncertainty associated with them. This view is especially important for operators to ensure that the constructed plans are accomplishing the higher-level campaign goals set out by the mission, and to what degree. It also highlights the impact of the goal changes for the next flyby (e.g., the addition of a new goal to observe plume X) at the strategic level (see the two bar charts on the bottom, with the impact gap in green). Furthermore, the operators are also presented with a few recommendations on how improve the plan to avoid conflicts and aid in campaign success. These recommendations are essential in fitting in with the iterative workflow of plan development.

Figure 10. Mission Impact tool: shows an overview of the simulations and the impact of updated or newly added goals to the mission progress

Downlink analysis

Subsystem Downlink Analysis Tool—The subsystem downlink analysis tool (shown in Figure 11) allows operators to review actual telemetry from downlink and compare it to modeled predicted data (predicts) to support the analysis of onboard health and safety and anomaly detection[16]. The tool resembles conventional downlink analysis tools, plotting onboard state over time overlaid with events, and a list of EVRs. The predicts are *clustered*, since the uplink process for autonomy produces thousands of predicts instead of just one. Operators can filter down to the cluster that match most closely what happened onboard in order to confirm that the spacecraft performed as expected, and identify improvements that they may need to make to the models. They can also filter EVRs to show exclusively unexpected EVRs, i.e., actual EVRs that did not match predicted EVRs for that plan cluster, or predicted EVRs that did not occur on the spacecraft. The tool can also display *inferred* data, i.e., the probabilistic distribution of key *unmeasured* states and parameters, reconstructed on the basis of received telemetry and of models of the spacecraft and its environment (the creation of such data products is discussed in detail in Section 5). Inferred data can help the operator explain *why* certain autonomy decisions were taken; the comparison of inferred data with predicts can also help surface the most likely explanation for unexpected outcomes, which they can use as a starting point for an investigation. The Subsystem Downlink Analysis tool provides a simple view of the subsystem performance for nominal cases. In off-nominal or poorly understood cases, operators can access more details in the Behavior Performance Analysis Tool, presented next.

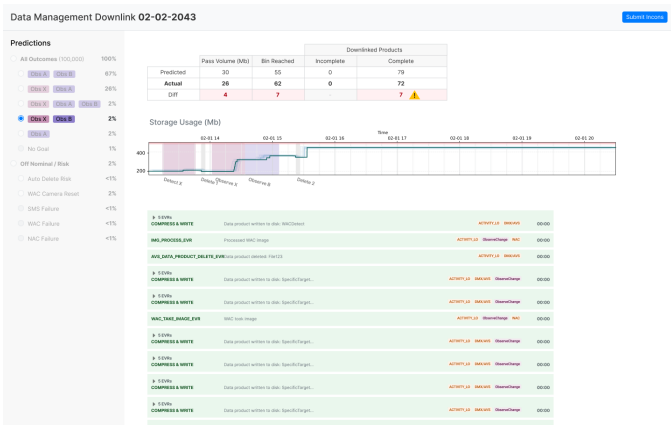

Figure 11. Subsystem Downlink Analysis Tool: allows downlink operators to compare predicts and actuals, and helps assess *what* happened onboard.

Behavior Performance Analysis Tool— The Behavior Performance Analysis Tool (Figure 12) allows autonomy experts and operators to understand the onboard decisions, and compare them with expected behavior, in order to debug unexpected outcomes and evaluate the performance of the autonomy. It repurposes the Task Network tool but, instead of a modeled preview, it presents downlink data juxtaposed with the thousands of predicts mentioned in the Subsystem Downlink Analysis tool. In support of this investigation, the tool also features messages describing onboard decisions, including (when applicable) previews of image data that were used to make decisions.

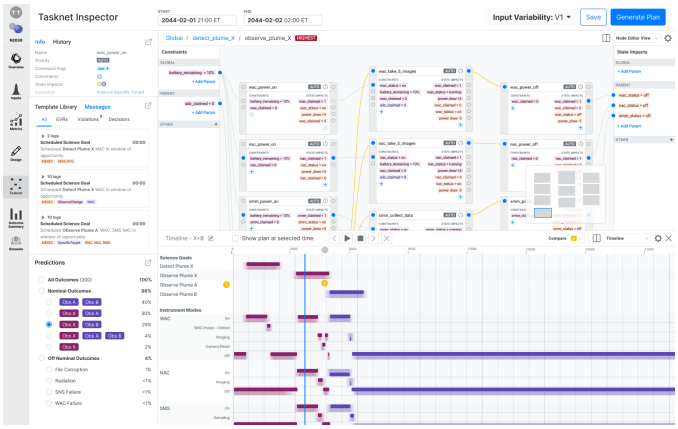

Figure 12. Behavior Performance Analysis Tool: shows the task network executed onboard, juxtaposed with predicts and with actuals, to support explanation of autonomy decisions.

5. Software Tools

New data-processing tools are needed to support the UI tools presented in the previous sections. In particular, we developed a performant simulation environment; a prediction engine that leverages the simulation environment to generate predicts; and an inference engine that uses telemetry data and spacecraft models to estimate the spacecraft's state, including, critically, the state of unmeasured or infrequently-measured variables.

Simulation Environment

We developed a simulation environment to simulate the behavior of the autonomous spacecraft, its surrounding environment, and the onboard autonomy. The simulation environment captures the spacecraft's position and attitude, thermal state, available power and energy, and available onboard storage; reproduces the behavior of the onboard instruments, namely, the wide-angle and narrow-angle cameras, the spectrometer, and the particles and plasma instrument; simulates the environment surrounding the spacecraft, in particular, the magnetic field's variability and the presence of plumes on Triton; and integrates with the MEXEC planning and execution software [4] and with a notional plume detector. The simulator also interacts with a ground data system by accepting input goals for the onboard autonomy and returning simulated telemetry, providing a *testbed* for the tools described in the rest of this paper. The simulation environment uses SPICE [17] for astrodynamics and planetary ephemerides, and ROS [18] for inter-process communications. Figure 13 shows the architecture of the simulation environment.

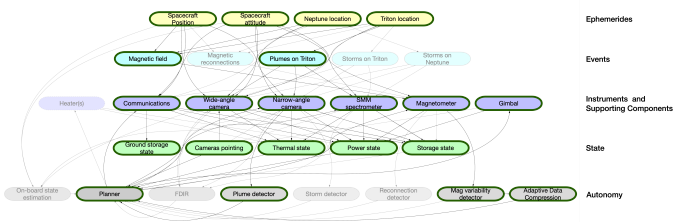

Figure 13. Software architecture of the proposed simulation environment. Modules circled in green are implemented; the remaining modules are slated for implementation in future work.

Prediction Engine

The high-level approach to predict outcomes consists of 3 main steps, namely, (i) sampling from the variability distributions to construct an input to the simulator, (ii) running the spacecraft simulator with the sampled input, and (iii) recording the various states and outputs in a shared database.

We use a Monte Carlo simulation approach, drawing on experience from the probabilistic prediction methodology used in the Copilot system [3] for predicting and visualizing the performance of M2020 rover execution with the Simple Planner autonomy. We extend this methodology to consider not only uncertainty with respect to activity's execution run time (e.g., delays) but also off-nominal scenarios and variations in event resource utilization (e.g., power, energy, thermal, and data storage).

While the current sampling method uses a simple Monte-Carlo [19] approach, future research includes testing more efficient sampling methods such as Latin Hypercube Sampling [20].

Implementation of the prediction engine presents two key difficulties. First, each simulation run needs to simulate a plan that spans many hours; while the simulation can run faster than real-time, there is a limit to the speed-up that can be achieved in simulation. Additionally, the prediction engine needs to be able to support simulating a wide range of runs, from a handful to hundreds of thousands of runs. As such, we must include parallelism and ability to configure resources based on the demand.

In order to overcome these difficulties, we built the prediction engine on Kubernetes, an open-source container orchestrator [21]. Not only does Kubernetes enable efficient orchestration and dynamic scaling, but it also easily allows deployment of the application on a cloud provider like Amazon Web Services (AWS) [22], to help meet the large computing power requirements. In order to achieve parallelism, we utilized the generator-worker method. In this method, the generator is responsible for generating input requests by sampling from the given variability distributions and pushing each input request onto a queue. Meanwhile, the worker takes an input request from the queue and runs a simulation based on the input, while storing data to a shared database. Figure 14 shows the workflow of the prediction engine, from the sampling to the storing of outcomes.

Inference Tool: Filling in the gaps

A key challenge in downlink analysis is to reconstruct the spacecraft's state and the decisions made by autonomy. While time-series data, EVRs, and engineering data products can provide a detailed view of the spacecraft's state and of the autonomy's decisions, it is generally non-trivial to correlate information across multiple time series and EVRs, and to reconstruct *why* autonomy made its decisions (that is, what elements of information influenced the autonomy's decisions). In addition, the amount of bandwidth available for engineering data is generally limited, since such information is in direct competition with scientific data products for limited downlink opportunities; this compounds the difficulty of reconstructing the spacecraft's state and its decisions, since data-intensive tools are often impractical (e.g., it may be infeasible to downlink the entire spacecraft state every time a planning algorithm is executed).

In order to address the twin challenges of (i) highlighting

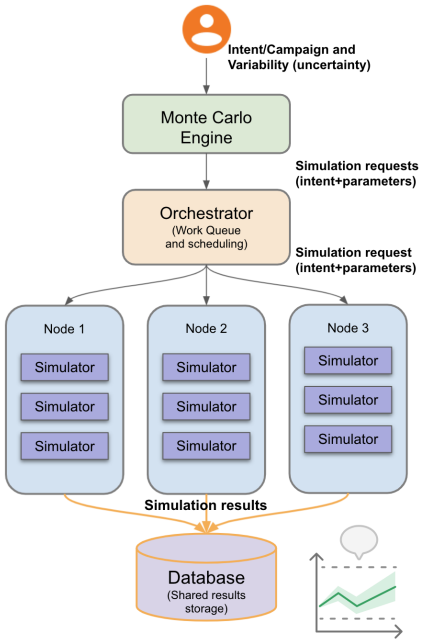

Figure 14. Overview of the architecture of the Prediction Engine.

correlations in the data sent by the spacecraft, and (ii) "filling in the gaps" wherever information is not downlinked or downlinked at lower-than-desired frequency, we advocate for the use of state estimation and inference algorithms. Such algorithms make use of *models* of the spacecraft, its environment, and, crucially, the onboard autonomy; based on these models and on the downlinked data, the algorithms reconstruct the *joint probability distribution* of the spacecraft's onboard state over time, enabling operators to assess the state of the spacecraft, and providing critical insight required to explain the autonomy's decisions.

As an added benefit, inference can reconstruct the state of environment variables that are not directly measured or are imperfectly sensed by the spacecraft; and, by exploiting models of the event, inference can help identify possible failures that may be otherwise masked by autonomy. Consider, for instance, an autonomous spacecraft tasked with detecting plumes on Triton and, if a plume is detected, perform follow-up investigations. How should an operator interpret a negative detection denoting the absence of a plume? The naive approach of downlinking all the image data supporting the autonomy's decisions is generally infeasible due to bandwidth constraints; in contrast, inference algorithms can exploit models of the detection algorithm (specifically, of its false positive and false negative rate) and of the expected event of interest (namely, the expected frequency of plume observations), and telemetry data collected across multiple observation opportunities, to assess the relative likelihood of multiple hypotheses, ranging from the absence of plumes to possible faults (e.g., the detection algorithm's threshold is too high, or the sensor gain is improperly set).

Modeling—Models of the spacecraft, of its environment, and of the onboard autonomy are needed to perform inference. In order to build such models in a principled way, we adopt the State Analysis framework [23], and employ state effect diagrams to represent the spacecraft state, and goal elaborations diagrams for the onboard autonomy. In this section, we provide a succinct description of state effect and goal elaboration diagrams; we refer the reader to [23] for a thorough discussion.

State effect diagrams capture the causal relationship between the states of the spacecraft and of its environment in a principled way. A state effect diagram provides a graphical representation of an analytical model for the relationship between individual states ("state effect model"); as such, it reduces the problem of capturing the coupled dynamics of a spacecraft and its environment into the much simpler problem of capturing how small subsets of states influence one another.

Figure 15 shows a state effect diagram for a spacecraft capturing magnetometry data, highlighting how the modeling problem is broken down into a number of much smaller, and more manageable, subproblems. In addition, the graph structure of the resulting model is highly amenable to computationally efficient inference, as discussed in the next section.

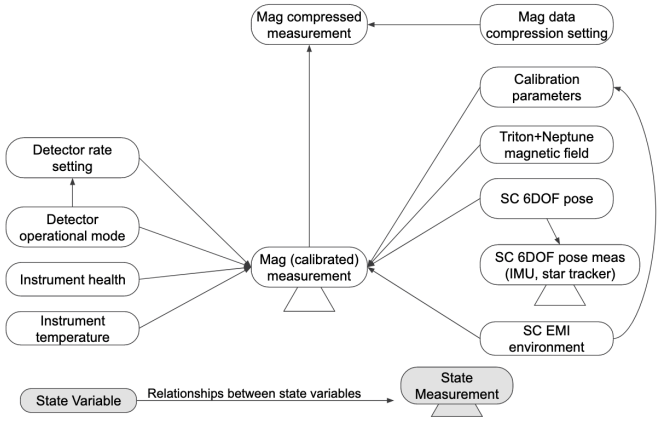

Figure 15. State effect diagram for a spacecraft capturing magnetometry data. State effect diagrams capture the topology of causal relationships between states, simplifying the modeling process and providing models that are amenable to efficient inference

Goal elaboration diagrams capture relationships between goals in the onboard autonomy. By providing an abstract, synthetic representation of the autonomy's goals, goal elaboration diagrams allow "peeking into" the decisions of autonomy, capturing the reasoning behind autonomy decisions (which makes such tools especially useful for explanation) while abstracting away algorithm-specific implementation details.

Goal elaboration diagrams focus on expressing the relationship between goals and actions - as such, they are well-suited for goal-based planners which explicitly optimize for achievement of user-specified goals. In contrast, this approach can struggle to capture the behavior of heuristics-based planners, where goals are imperfectly mapped to decisions through heuristics that cannot be properly modeled in a causal fashion; in such cases, a black-box representation of autonomy (using the exact same code employed onboard the spacecraft) may be used, resulting in increased fidelity but much lower interpretability.

Figure 16 shows a goal elaboration diagram for an autonomy module autonomously adapting the sampling rate of a mag-

netometer in response to its measurements.

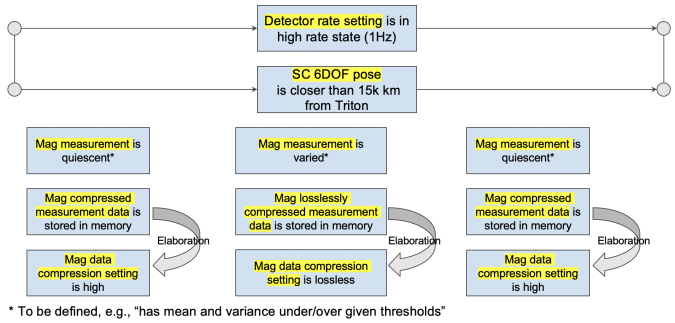

Figure 16. Goal elaboration diagram for an instrument autonomously adapting its sampling rate in response to the measurements. Goal elaboration diagrams capture the relationship between the autonomy's goals and commanded actions in a systematic way, capturing the reasoning behind autonomy decisions while abstracting away specific implementation details.

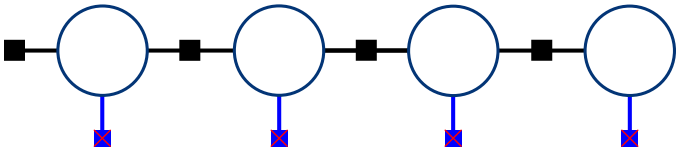

Figure 17. An example of a simple multi-modal factor graph used for inference. Here we see magnetic field strength variables we are estimating as circles, connected in time by black nodes representing how magnetic field strength is expected to change with time. The blue nodes connected to each variable represent measurements of the magnetic field from the spacecraft and their associated uncertainty. These factors are multi-modal because they are *detachable*; MH-ISAM2 considers two hypothesis for each of these factors and evaluates the respective factor graphs accordingly. The hypothesis with the least accumulated error is considered the most likely.

Algorithms—In order to perform inference, we use state effect and goal elaboration diagrams to build a factor graph [24] representation of the spacecraft. Like Bayes nets, factor graphs represent the joint probability distribution of a set of states as a product of factors; factors, in turn, can be specified as any function over a subset of state variables (for instance, the causal relationship between a spacecraft's attitude and the likelihood of observing a given region of a body can be represented as a factor, where the inputs are the spacecraft's attitude states, and the output is the likelihood that the spacecraft's camera points towards the region of interest). Unary factors are also possible, where a factor is a function over a single variable (for example, a direct measurement of a variable can be added to the graph as a unary factor, also called a *prior*). We adopted the open source library GTSAM [25] to construct new types of factors based on the spacecraft's continuous measurements and on the variables we would like to infer. The factor graph is then solved as a nonlinear optimization problem, where we determine the set of variables that best explain the measurements. GTSAM does this efficiently by exploiting sparsity (i.e., the fact that factors typically only affect a small portion of the total number of variables).

To handle discrete variables and measurements, we make use of a version of GTSAM called MH-ISAM2 [26] which extends GTSAM by adding multi-modal factors. Multi-modal factors can be used to represent discrete sets of states or models (e.g., a plume exists or doesn't exist, the observed magnetic field conforms to one proposed model or another, one of the sensors is operating normally or has experienced a fault) as separate hypotheses; factor graph optimization then finds the hypothesis which best explains the measurements. An example of one type of multi-modal factor is shown in Figure 17, where circles represent the variable of interest (in this case the magnetic field strength) that we are trying to infer at sequential time steps. Black nodes connecting the variables represent factors, which have a probability distribution associated with them. In this example, the factors connecting sequential variables describe how models say the magnetic field strength should progress with time. The blue factors are so-called *detachable* unary factors that represent magnetometer measurements, with an associated measurement noise uncertainty. If one of the magnetometer measurements is faulty (i.e., the model of the magnetic field says that the magnetic field strength is unlikely to abruptly change in time, but measurement shows a large temporary change in magnetic field strength that can't be explained by measurement noise), the factor graph is able to consider this option as a separate hypothesis, where the measurement is detached from the graph optimization problem. The set of hypothesis that result in the least accumulated error between the estimated states and their associated factors is then considered the most likely explanation.

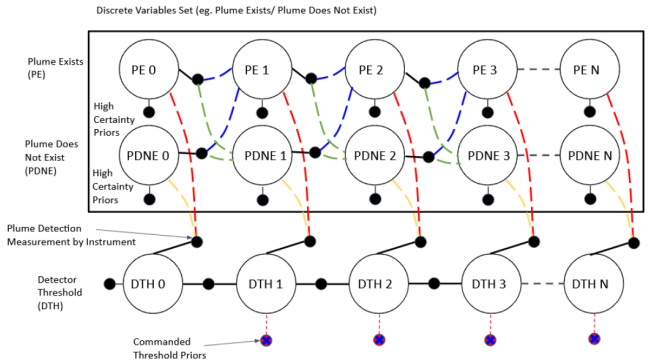

Figure 18. An example of a multi-modal factor graph with discrete variables. We model discrete variables, in this example plume existence, as two separate sets of variables. These are then both connected to the detector threshold variable through *multi-association* factors representing the hypothesis that the plume exists, or does not exist.

Another use case for multi-modal factor graphs are discrete variables. One example of interest is the problem of assessing the existence of a plume on Triton, and the ability to detect plumes with a camera, as shown in Figure 18. To model this problem, we group together two parallel sets of variables in time, one where a plume exists, and one where it does not, and connect these to a threshold variable representing the properties of the detector through *multi-association* factors. The multi-association factors represent the hypothesis that either mode of the discrete variable for plume existence explains the detection measurement. The discrete variables are also connected in time through additional multi-association factors that represent the likelihood of a plume transitioning between existing and not existing. We also consider the likelihood that the threshold variable we have commanded for the detection algorithm has not been applied correctly with detachable unary factors.

The output of the factor graph is a *maximum-likelihood estimate* of the state variables considered, and the *marginal distribution* of each variable. In future work, this data will be displayed in the Subsystem Downlink Analysis Tool (Figure 11), providing operators with key insight into unmeasured variables and, critically, with the likelihood of each considered hypothesis - helping operators assess the state of the spacecraft and understand why autonomy made its decisions.

6. USER STUDY

Study Design

In order to qualitatively evaluate the performance of tools and initial autonomous planning procedures, we conducted a light-weight simulation of future autonomous operations. The approach follows the idea of Design Simulation [27], a design method introduced at JPL to expand the scope of User Enactments [28], [29], a type of Experience Prototype [30] to assess the experience of operators, and to provide systems engineers with insight into how to improve the efficacy of their operations early in the mission design process. The immersive effect of Design Simulations provides planners and operators with an opportunity to experience future operations concepts and software early in a mission life cycle, and use the feedback from these simulations to inform planning conversations before final tools are put into practice during flight. In this work, we took an experimental approach in order to gather recommendations that could scale to future missions utilizing autonomous technologies. We developed a loose impression of our design principles for operational processes and tools, rather than a specific concept of operations. This allowed us to focus on a scalable framework that future missions can use as a starting point.

The user study needed to compare the strong prior that operators bring to how they understand conventional operations, against how future autonomy-based missions might deviate from a more deterministic set of practices. Our early formative research of operations practices revealed a number of rich areas to investigate the potential impact of this newly-experienced autonomy. In particular, we chose to focus on how scientists would react to probabilistic resource conflicts, whether operators would trust and accept a non-deterministic uplink plan, and whether operators would feel confident in their retrospective reconstruction of onboard behavior and safety upon downlink.

To study these particular facets of operations, we selected the previously described "Mapping Triton and Plume Detection" scenario (see Section 2), and elaborated preliminary tools and an operations context that fit this scenario. The user study introduced operational issues that the participating operators would have to resolve using the provided tools. The study began by presenting operators with telemetry that identified a new, unexpected plume from the previous downlink. We guided the participants to explore follow-up observations around this interesting new feature, which created a conflict with the baseline plan, and included no guarantee that it would even still exist during that planning window. Then we guided them to create a plan that would only image the new feature on the condition that it was determined by the onboard detection algorithm to still exist.

As the test proceeded, after uplink, the spacecraft confirmed the existence of the transient feature and imaged it. However, during data collection, we injected a camera reset into the scenario, that only allowed the spacecraft to successfully image the lower priority of the pre-existing features. We used this unexpected spacecraft behavior as a way to evaluate operator trust in the autonomous system.

Participants

We recruited six participants, including two who were already familiar with the scenario, and four practitioners from flight projects at JPL. We introduced them to the proposed mission concept, the onboard autonomy that the spacecraft would be using to determine its actions, and the suite of tools that we created for them to use to command the spacecraft. We then walked them through an overview of the user study procedure, which we describe below.

Procedure

Our study consisted of three main sections over the course of three days that represented a loose operations concept: goal identification and design, high fidelity simulations and uplink, and downlink analysis. The participants role-played the operations positions in conversations and decision-making across those meetings, though we also encouraged them to break into reflective discussions about the process and tools along the way. We also collected feedback in the form of survey responses in a journal they used for each day. We had implemented the tools as click-through prototypes only, and user study facilitator needed to "drive" the tools at the the operator's request. This dynamic evoked dialogue about what information they needed to see and why.

Analysis

Once we concluded the study, we transcribed the audio from the sessions and compiled the survey responses. To identify themes, we grouped clusters of related observations into affinity groups, focusing on topics that related to our research questions.

Findings

In this section, we describe preliminary themes that emerged from our analysis. The user study participants in general successfully used the tools and series of steps to complete their operations tasks. The operators' participation, inquiries, and suggestions highlighted successes and opportunities to improve our first pass at an autonomous operations experience. Participants' behavior also revealed how negotiation dynamics might change in a highly modeled autonomous paradigm.

Participants' requests and expectations exposed that the system lacked detailed data on particular screens that our preliminary design hadn't anticipated. For example, we had designed the system to include progressive disclosure of higher fidelity science goal prediction details. These predictions indicated the probability of each goal being successfully executed, given the order of operations and science targets. Lower-fidelity predictions, which we made available to participants at the activity start, gave operators a rough estimate of the outcome of the plan, while higher fidelity predictions, which we made available later in the activity, provided estimations based on a Monte Carlo Simulation approach that modeled the outcomes across 10,000 cases.

While participants generally accepted the progression from low fidelity to high fidelity data, conversations with scientist participants revealed that some wanted higher fidelity data available at the beginning of negotiations. In this particular situation, the decision about which observations to prioritize

kicked off a lively discussion about the various possible outcomes and their combinations. During the later high fidelity phase scientists focused on the likelihood that the spacecraft might encounter all the plumes of interest. This favorable (and with a 2% likelihood, very unlikely) simulation outcome, where the spacecraft delivered all the desired science across plumes of interest, hid in the later high fidelity outcomes. As one scientist described, "...starting with the premise that we cannot do all 3 [observations], spending a month of arguing about which way to go for the plan, to find out there is a chance to do all 3 that could have led to a lot of arguments and drama for nothing." As a takeaway, this observation opens a space for earlier views to include more Monte Carlo Simulations to facilitate earlier negotiations, both to be used in discussions, and as a starting point for trying to improve favorable outcomes.

In order to discuss what might happen onboard, operators used the prototype tools to cross-compare possible outcomes, represented visually as side-by-side resource and task timelines. They could toggle between clusters of the predicted, related, sequential plans. We observed participants using the summary of all the possible outcomes as a starting point, using it to identify cases where outcomes came close to the resource limits, or where certain goals happened at different times. With this overview knowledge, they could then zoom into the details of specific plan sequences to understand the range of details and conditions defining how that plan might have occurred. For example, one scientist pointed out "[An] advantage that comes out of being able to toggle between them [summary and specifics] is that it gives you hints "I'm running into [a] time [boundary issue] here, or I'm hitting the power limit". Those pieces can help people figure out... if they want to ... change the situation.". The autonomy engineer agreed, adding "One of the big things we would do in a meeting like this is look at the different timelines... We should review and check "ok everybody, the 2 most likely times for A are here and here, are they both good?" We interpret this to mean that the ability to access and toggle between possible and high-probability outcomes facilitated an understanding of particular outcomes of a variety of plans against a comprehensive space of possible outcomes. Additional work will need to be done to develop meaningful clusters that can facilitate exploration of more complicated plans, each with many interacting onboard goals, and upwards of 20 related plan clusters.

In a process that followed patterns from conventional ground operations, we also observed downlink operators taking advantage of the "predicts vs actuals" view provided by the Subsystem Downlink Analysis tool. In order to build an understanding of what transpired onboard the spacecraft, operators inspected differences between the predicts that matched onboard goal profiles, and actuals, to assess performance and identify unexpected onboard behaviors. In our study, operators could perform this comparison using different plan clusters. The operators identified the potential for this to help understand cases where data about the onboard execution is not available or clearly reconstructable, e.g., in instances such as data loss. One spacecraft engineer explained, "[if] we don't downlink that data, we wouldn't know how the plan evolved over time; if we only have EVRs and EHA, then the ability on the downlink dashboard to cycle between the different models could give us a sense of what the onboard planner chose, before we downlink that data and learn the real answer."

Future software tools based on the prototypes of this study should reflect the complexity of the behavior and operational needs for a more robust mission concept. During the study, the operators focused mostly on the timeline visualization and less on the graph task network view. The autonomy engineer elaborated: "I did almost all my analysis using the Gantt/timeline chart and the little warning or error messages, and looked at the resource plot to do a sanity check on the error messages." We believe that while simpler, well understood plans and behaviors (such as the one used in our study) can leverage timeline views, more complex plans with poorly understood spacecraft behavior might require additional views to supplement the timeline, such as the network block diagram in the Behavior Performance Analysis Tool.

We observed that the use of models and probabilistic results influenced the discussion of science priorities. In one example, we observed that, when one scientist added a new autonomy-enabled observation goal, it reduced the likelihood of their own observation from almost 100% to 70%. The impacted scientist felt that the changes had compromised their original observation, even thought it was still likely to occur, and, as a result, the participants discussed several strategies to make a compromise or otherwise relax the plan without using autonomy. The impacted scientist explained "I would have pushed harder if there was any indication that [the baseline plumes] or both were super important and we were not able to image them again." At the same time, some scientists noted that, while they had to make decisions based on predictive models, those models attempted to characterize parts of the Solar System that humans don't know very much about. Some scientists described this as somewhat arbitrary and even potentially contentious. While the mission planner responded with equanimity, observing "Such is planning", the autonomy engineer suggested that perceived or actual arbitrariness could be refined throughout the ongoing mission.

Finally, we observed a change in how science teams prioritize observations and discuss their likelihood. While sequence-based operations teams have to cull observations from the plan using their implicit understanding of the tradeoffs, our operations team front-loaded their plan with desired science observations, ranked them by priority and debated their likelihood using clusters of possible outcomes from high fidelity simulations. As the autonomy engineer commented, "It used to be a discussion or argument about what is more important to fit in and what's in and out. With this [autonomous] planning paradigm it's a discussion of ranking the activities because we're never really sure what's in and out, we know the highest priority things will almost certainly happen and the lowest will almost certainly not; we won't know where the cutoff is until we run the model, and really, until we run it on the spacecraft."

7. CONCLUSIONS

In this paper, we investigated the problem of *operations for autonomy*, i.e., of identifying the tools and workflows that will be required to effectively operate future highly autonomous spacecraft. We grounded our analysis in a notional flyby mission to the Neptune-Triton system, a destination selected because the resulting low bandwidth and high latency makes it attractive for deployment of autonomy, and also results in highly challenging operations. After identifying a set of autonomy-enabled science scenarios that are likely to challenge current operations paradigms, we identified how operations workflows and roles are likely to evolve to

accommodate onboard autonomy. We then presented the preliminary design of user interfaces and software tools that respond to the needs of future autonomy workflows; the tools were evaluated in a user study with JPL mission operators, resulting in broadly favorable findings.

The work in this paper paves the way for a number of interesting directions for future research. First, we plan to directly address the operators' suggestions that emerged in the user study by (i) increasing the fidelity of initial predictions presented to the users (bringing the some of the benefits of the high fidelity simulations sooner in the process) and (ii) developing clustering tools that support a large number of possible outcomes. Second, we plan to continue the integration of inference and UX tools, helping operators interact with inferred data to assess the spacecraft's state and the autonomy's decisions. This will include the design of tools for uplink explanation and advising. Third, we plan to perform follow-on user studies on scenarios more representative of real missions' operational needs, including larger and more complex task networks, multiple conflicting science objectives, and multiple uplink-downlink cycles. Finally, we plan to continue the development and integration of the prototypes presented in this paper with the long-term goal of infusing the findings of this work in future ground data systems, in particular AMMOS [31], to support future autonomous space exploration missions.

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Table 1. Uplink and downlink operations capabilities required by the mission scenarios

Scenario		Modeling intent												Prediction							Visualization		Explanation					Advising	
		Campaign representation with high level goals	Campaign representation with traditional command sequences	Campaign representation w. high level goals *and* command sequences	Hierarchical plan specification	Conditional observations	Repeated observations	Science goals prioritization	Geometry constraints	Resource constraints	Multiple campaigns with conflicting goals/resources	Represent parameters for FDIR and detection	Intent not achievable (contraint violation)	Intent close to safety thresholds (brittle cases)	Execution Timing prediction	Data Volume prediction	Power Usage prediction	Off-nominal/degradation prediction	State uncertainty prediction (e.g. pointing)	Summarization of prediction	Play back particular plan execution	Constraint violation analysis	Task priorities analysis	Input parameters analysis	Reasons to why something did (not) work	Autonomy behavior analysis.	Constraint relaxation	Goal relaxation
Monitoring	S1: data prioritization		✓				✓			✓						✓				✓								
	S2a: Event detection, resource neutral	✓				✓				✓	✓					✓	✓			✓								
	S2b: Event detection and replanning	✓				✓	✓			✓	✓					✓	✓			✓			✓			✓	✓	✓
Opportunistic observations during mapping	S1a, plume detection: missed			✓		✓		✓								✓				✓								
	S1b, plume detection: interrupt			✓	✓	✓		✓	✓	✓	✓					✓				✓								
	S1c, plume detection: coexist			✓	✓	✓		✓	✓	✓	✓					✓				✓								
	S2: FDIR during mapping	✓					✓					✓			✓	✓		✓		✓	✓				✓	✓	✓	
	S3a, storm detection: missed	✓				✓	✓	✓		✓						✓	✓			✓								
	S3b, storm detection: interrupt	✓				✓	✓	✓	✓	✓						✓	✓			✓	✓		✓			✓	✓	
	S3c, storm detection: coexist	✓				✓	✓	✓	✓	✓						✓	✓			✓								
Opportunistic observations during targeted observation	S1: off-nominal engine burn			✓			✓	✓		✓	✓	✓	✓			✓	✓	✓		✓	✓	✓						
	S2a: instrument parameters, resource neutral	✓			✓	✓	✓			✓		✓			✓	✓	✓	✓		✓			✓	✓	✓	✓	✓	✓
	S2a: instrument parameters, replan resource use	✓			✓	✓	✓			✓		✓			✓	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓	✓

Scenario		**Explain data filtering/compression**	**Explain event detection/no detection**	**Explain planner/scheduler decisions based on intent**			**Estimate and Propagate State**		**Explain FDIR activities**		
				Resource-aware: onboard state	Uncertainty-aware: Onboard state uncertainty	Geometry-aware: onboard state	Spacecraft state	Environment state	What triggered FDIR	What did FDIR think was happening	How did FDIR react
Monitoring	S1: data prioritization	✓						✓			
	S2a: Event detection, resource neutral	✓	✓								
	S2b: Event detection and replanning	✓	✓	✓							
Opportunistic observations during mapping	S1a, plume detection: missed		✓	✓		✓	✓	✓			
	S1b, plume detection: interrupt		✓	✓		✓	✓	✓			
	S1c, plume detection: coexist		✓	✓		✓	✓	✓			
	S2: FDIR during mapping								✓	✓	✓
	S3a, storm detection: missed		✓	✓		✓	✓	✓			
	S3b, storm detection: interrupt		✓	✓		✓	✓	✓			
	S3c, storm detection: coexist		✓	✓		✓	✓	✓			
Opportunistic observations during targeted observation	S1: off-nominal engine burn			✓		✓	✓		✓	✓	✓
	S2a: instrument parameters, resource neutral							✓	✓	✓	✓
	S2a: instrument parameters, replan resource use			✓				✓	✓	✓	✓

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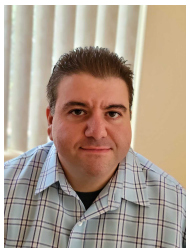
of icy materials, and thermodynamic and geophysical modeling of icy worlds and cometary environments to apply the experimental results.



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Michel Ingham is the Chief Technologist of JPL's Systems Engineering Division. Since he joined JPL in 2003, he has worked as a software systems engineer and system architect on a variety of projects, including the Mars Science Laboratory rover, and the Europa Clipper mission. He has led several NASA, JPL and DARPA research and development activities, in the areas of model-based systems and software engineering, software architectures, and spacecraft autonomy. Dr. Ingham received his Sc.D. and S.M. from MIT's Department of Aeronautics and Astronautics, and his B.Eng in Honours Mechanical Engineering from McGill University in Montreal, Canada.

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Iain Tierney is starting his senior year of his bachelor's degree at the University at Buffalo studying Computer Science. His professional interests include software development and engineering, web development, artificial intelligence, and cloud computing.